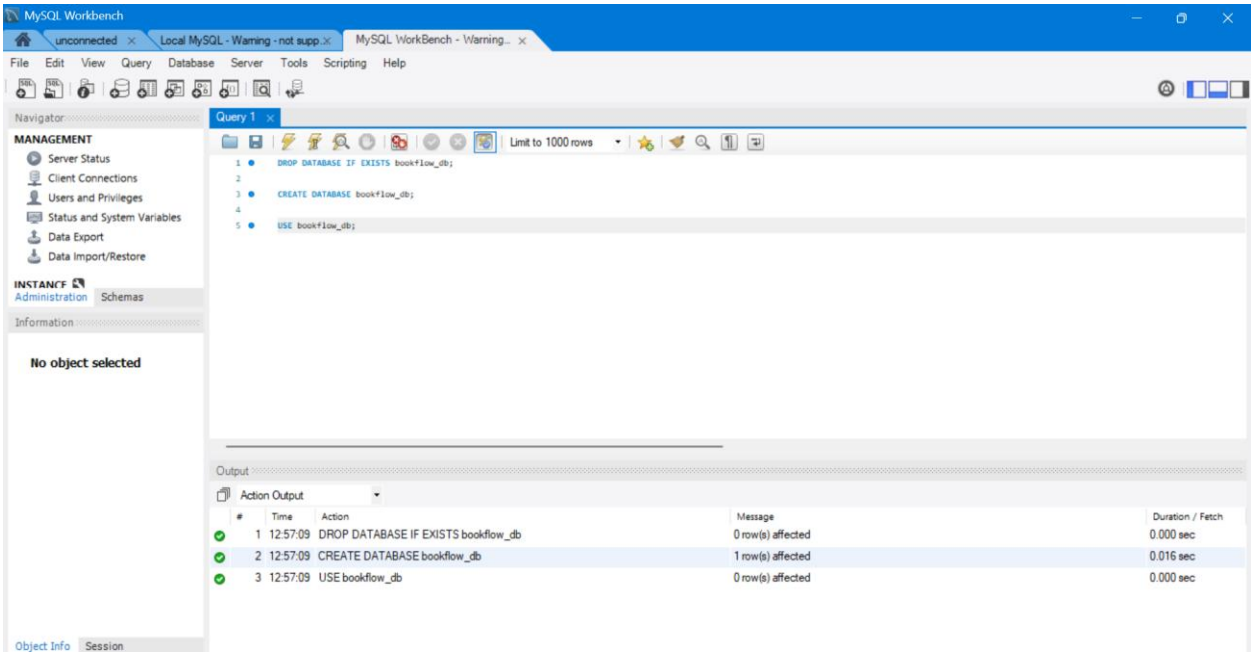
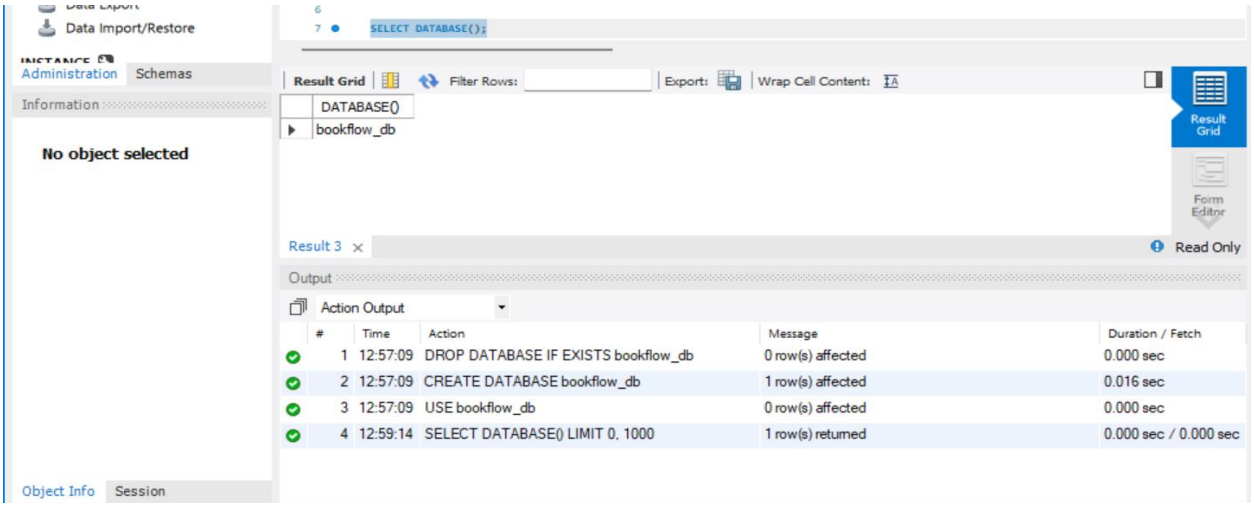
Database System Engineering and Distributed Backend Development

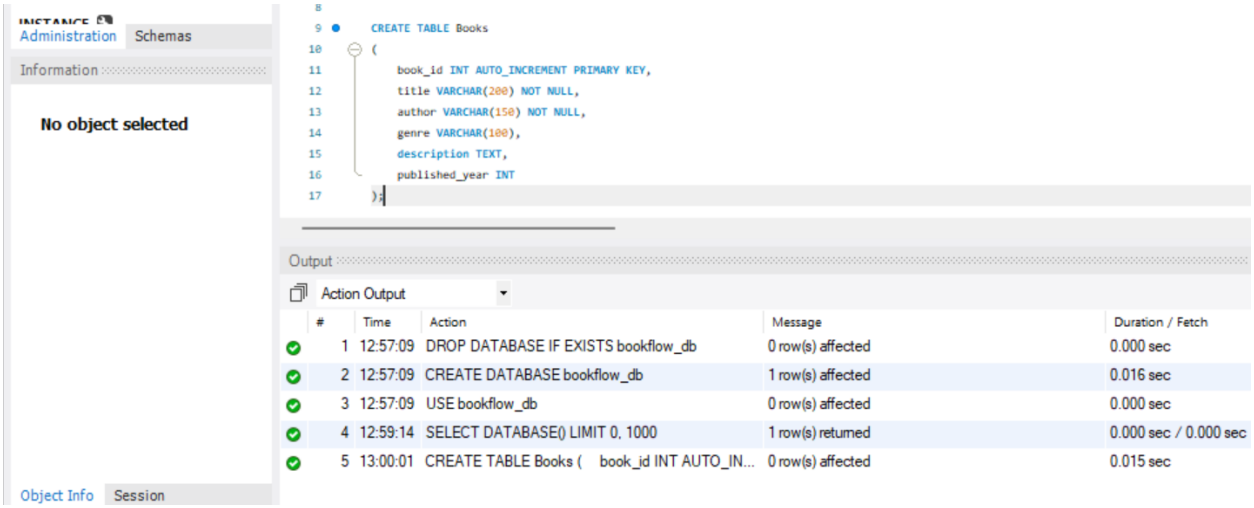
Course Code: 25CS1302E

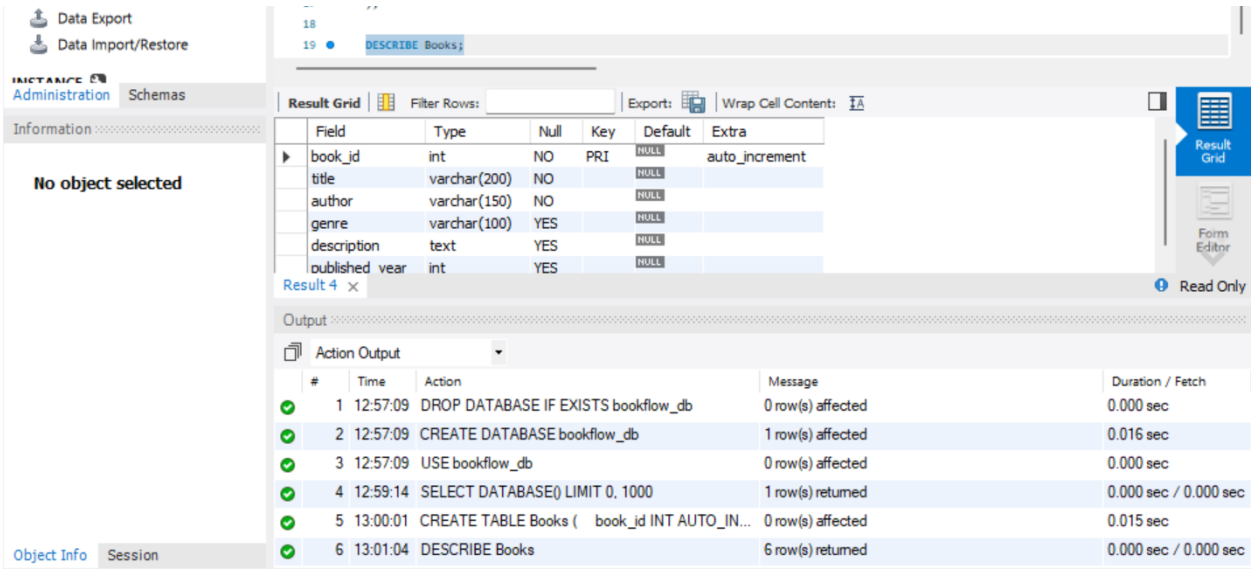
TOPIC: WEEK- 04

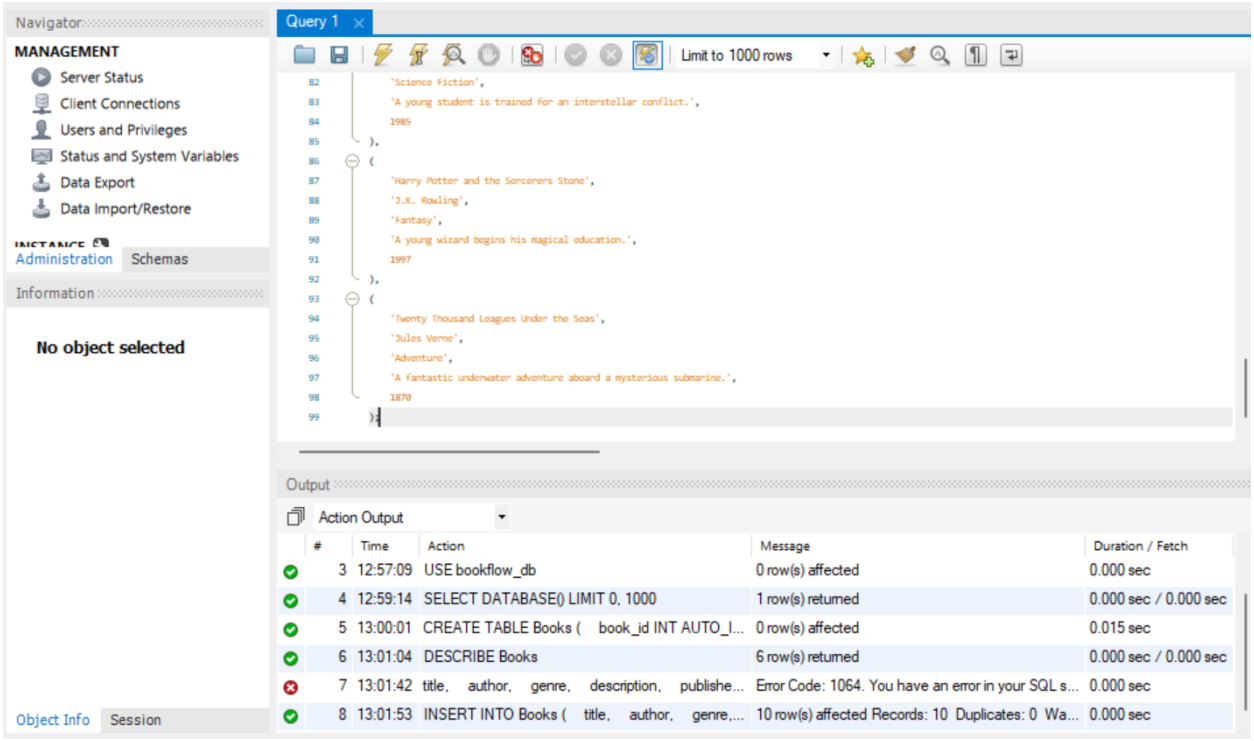
Name : S. Manvi Vaishnavi
Roll Number : 2520030326

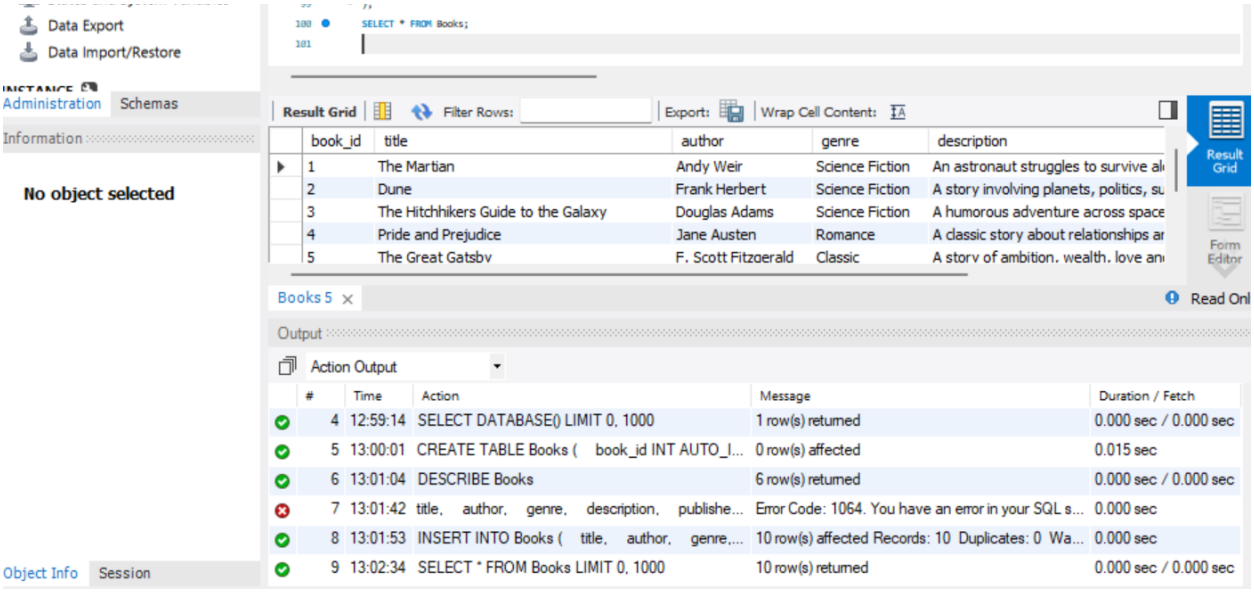

MySQL Workbench
unconnected
Local MySQL - Warning - not supp...
MySQL WorkBench - Warning...
File Edit View Query Database Server Tools Scripting Help
Navigator
MANAGEMENT
Server Status
Client Connections
Users and Privileges
Status and System Variables
Data Export
Data Import/Restore
INSTANCE
Administration Schemas
Information
No object selected
Object Info Session
Query 1
Limit to 1000 rows
1 DROP DATABASE IF EXISTS bookflow_db;
2
3 CREATE DATABASE bookflow_db;
4
5 USE bookflow_db;
Output
Action Output
Time Action Message Duration / Fetch
1 12:57:09 DROP DATABASE IF EXISTS bookflow_db 0 row(s) affected 0.000 sec
2 12:57:09 CREATE DATABASE bookflow_db 1 row(s) affected 0.016 sec
3 12:57:09 USE bookflow_db 0 row(s) affected 0.000 sec

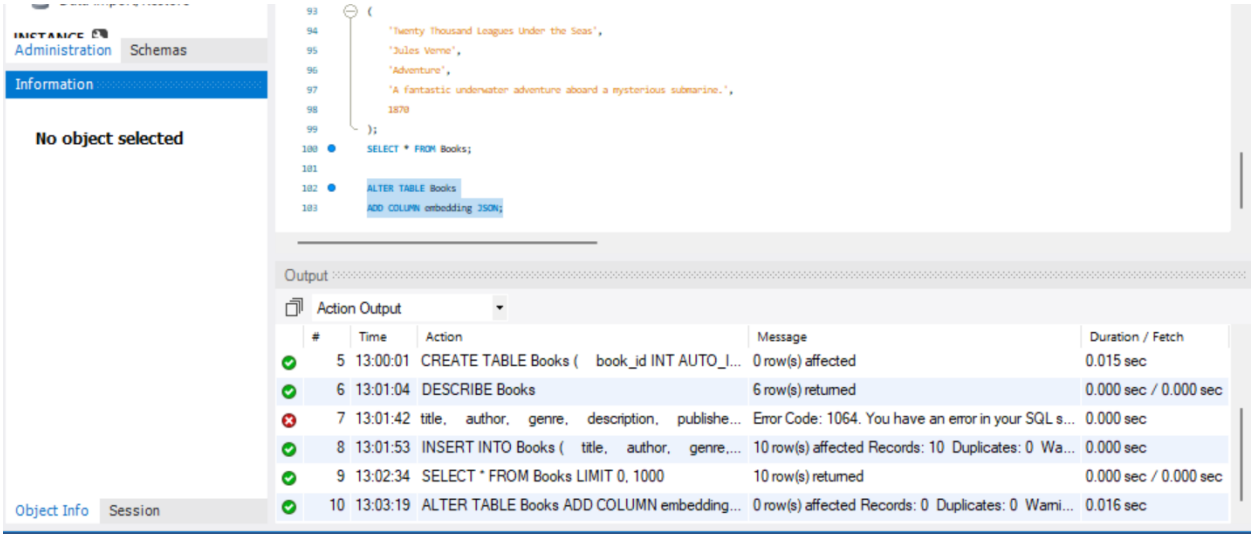

Data Import/Restore
Administration Schemas
Information
No object selected
Object Info Session
6
7 SELECT DATABASE();
Result Grid Filter Rows: Export: Wrap Cell Content:
DATABASE()
bookflow_db
Result Grid
Form Editor
Result 3
Read Only
Output
Action Output
Time Action Message Duration / Fetch
1 12:57:09 DROP DATABASE IF EXISTS bookflow_db 0 row(s) affected 0.000 sec
2 12:57:09 CREATE DATABASE bookflow_db 1 row(s) affected 0.016 sec
3 12:57:09 USE bookflow_db 0 row(s) affected 0.000 sec
4 12:59:14 SELECT DATABASE() LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec

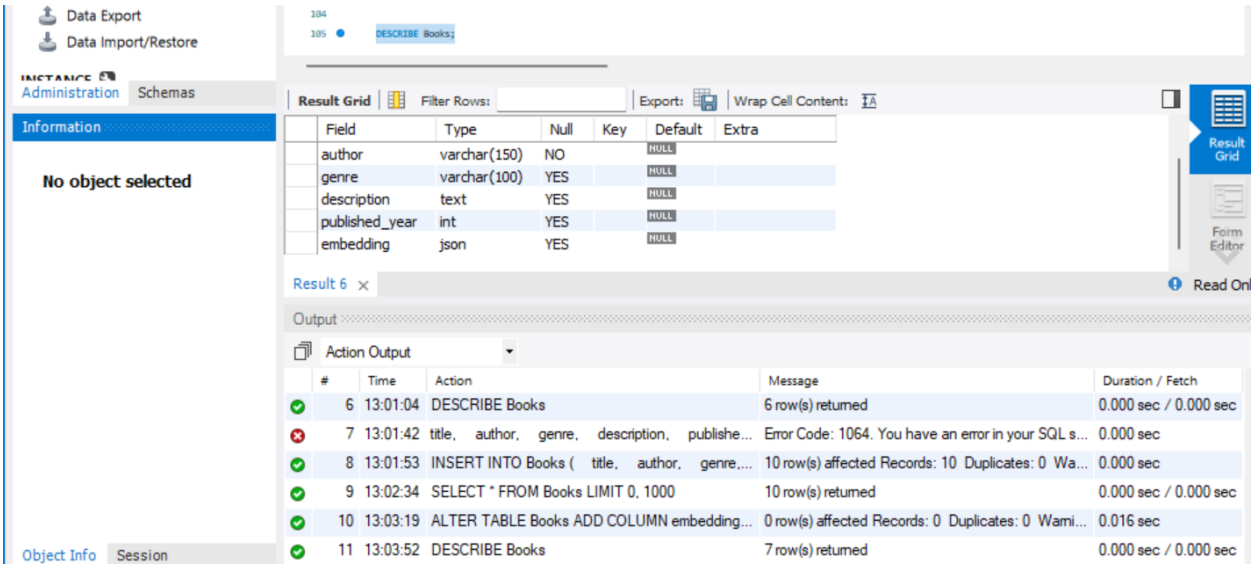

Administration
Schemas
Information
No object selected
Object Info
Session
CREATE TABLE Books
(
book_id INT AUTO_INCREMENT PRIMARY KEY,
title VARCHAR(200) NOT NULL,
author VARCHAR(150) NOT NULL,
genre VARCHAR(100),
description TEXT,
published_year INT
);
Output
Action Output
Time Action Message Duration / Fetch
1 12:57:09 DROP DATABASE IF EXISTS bookflow_db 0 row(s) affected 0.000 sec
2 12:57:09 CREATE DATABASE bookflow_db 1 row(s) affected 0.016 sec
3 12:57:09 USE bookflow_db 0 row(s) affected 0.000 sec
4 12:59:14 SELECT DATABASE() LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
5 13:00:01 CREATE TABLE Books (book_id INT AUTO_IN... 0 row(s) affected 0.015 sec

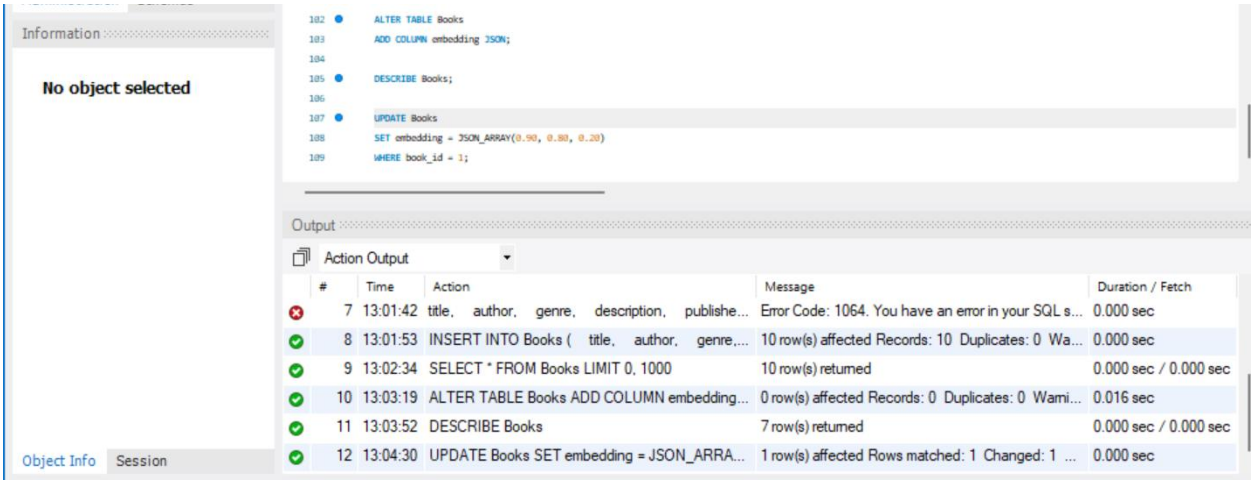

Data Export
Data Import/Restore
Administration
Schemas
Information
No object selected
Object Info
Session
DESCRIBE Books;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Field Type Null Key Default Extra
book_id int NO PRI NULL auto_increment
title varchar(200) NO NULL
author varchar(150) NO NULL
genre varchar(100) YES NULL
description text YES NULL
published_year int YES NULL
Result 4
Read Only
Output
Action Output
Time Action Message Duration / Fetch
1 12:57:09 DROP DATABASE IF EXISTS bookflow_db 0 row(s) affected 0.000 sec
2 12:57:09 CREATE DATABASE bookflow_db 1 row(s) affected 0.016 sec
3 12:57:09 USE bookflow_db 0 row(s) affected 0.000 sec
4 12:59:14 SELECT DATABASE() LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
5 13:00:01 CREATE TABLE Books (book_id INT AUTO_IN... 0 row(s) affected 0.015 sec
6 13:01:04 DESCRIBE Books 6 row(s) returned 0.000 sec / 0.000 sec

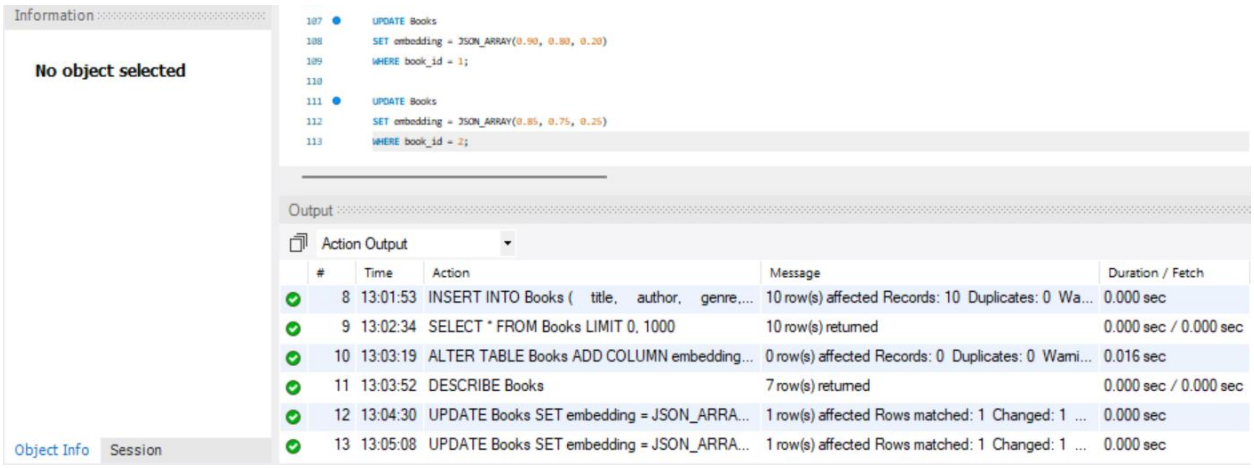

Navigator
MANAGEMENT
Server Status
Client Connections
Users and Privileges
Status and System Variables
Data Export
Data Import/Restore
Administration
Schemas
Information
No object selected
Object Info
Session
Query 1
Limit to 1000 rows
'Science Fiction',
'A young student is trained for an interstellar conflict.',
1985
),
(
'Harry Potter and the Sorcerers Stone',
'J.K. Rowling',
'Fantasy',
'A young wizard begins his magical education.',
1997
),
(
'Twenty Thousand Leagues Under the Seas',
'Jules Verne',
'Adventure',
'A fantastic underwater adventure aboard a mysterious submarine.',
1870
);
Output
Action Output
Time Action Message Duration / Fetch
3 12:57:09 USE bookflow_db 0 row(s) affected 0.000 sec
4 12:59:14 SELECT DATABASE() LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
5 13:00:01 CREATE TABLE Books (book_id INT AUTO_I... 0 row(s) affected 0.015 sec
6 13:01:04 DESCRIBE Books 6 row(s) returned 0.000 sec / 0.000 sec
7 13:01:42 title, author, genre, description, publishe... Error Code: 1064. You have an error in your SQL s... 0.000 sec
8 13:01:53 INSERT INTO Books (title, author, genre,... 10 row(s) affected Records: 10 Duplicates: 0 Wa... 0.000 sec

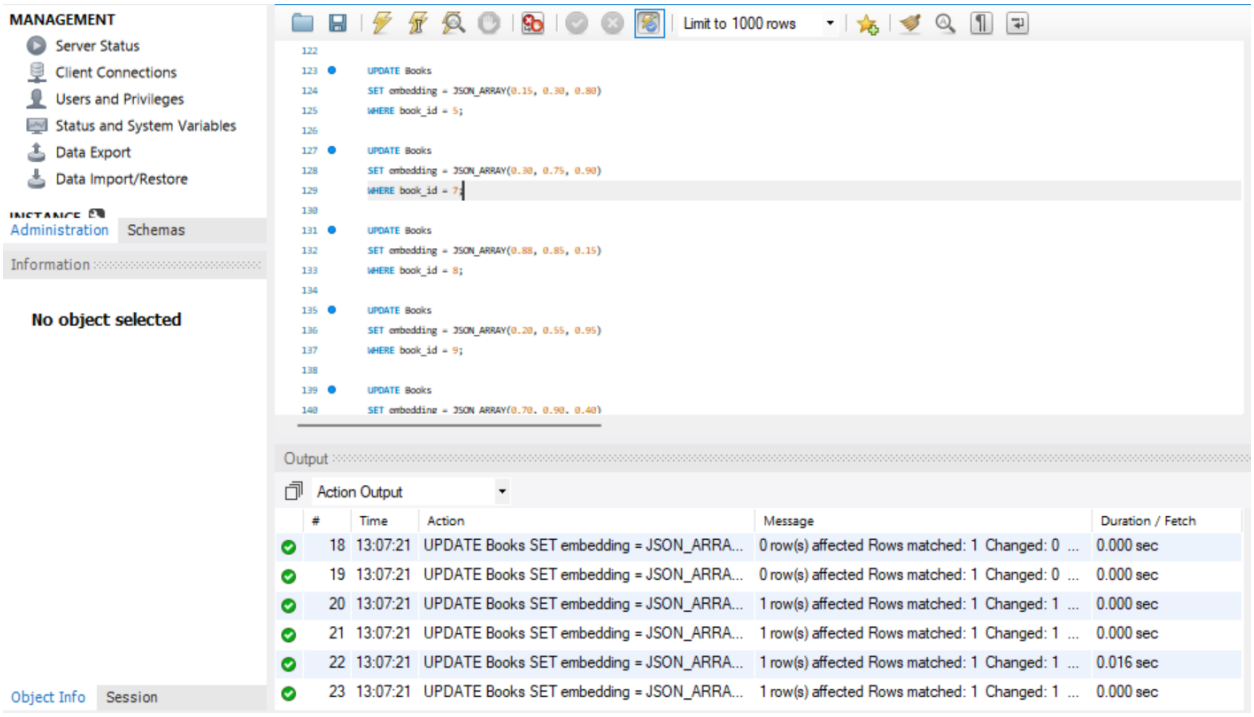

Data Export
Data Import/Restore
Administration
Schemas
Information
No object selected
Object Info
Session
SELECT * FROM Books;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
book_id title author genre description
1 The Martian Andy Weir Science Fiction An astronaut struggles to survive al
2 Dune Frank Herbert Science Fiction A story involving planets, politics, su
3 The Hitchhikers Guide to the Galaxy Douglas Adams Science Fiction A humorous adventure across space
4 Pride and Prejudice Jane Austen Romance A classic story about relationships ar
5 The Great Gatsby F. Scott Fitzgerald Classic A story of ambition, wealth, love an
Result Grid
Form Editor
Books 5
Read Onl
Output
Action Output
Time Action Message Duration / Fetch
4 12:59:14 SELECT DATABASE() LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
5 13:00:01 CREATE TABLE Books (book_id INT AUTO_I... 0 row(s) affected 0.015 sec
6 13:01:04 DESCRIBE Books 6 row(s) returned 0.000 sec / 0.000 sec
7 13:01:42 title, author, genre, description, publishe... Error Code: 1064. You have an error in your SQL s... 0.000 sec
8 13:01:53 INSERT INTO Books (title, author, genre,... 10 row(s) affected Records: 10 Duplicates: 0 Wa... 0.000 sec
9 13:02:34 SELECT * FROM Books LIMIT 0, 1000 10 row(s) returned 0.000 sec / 0.000 sec

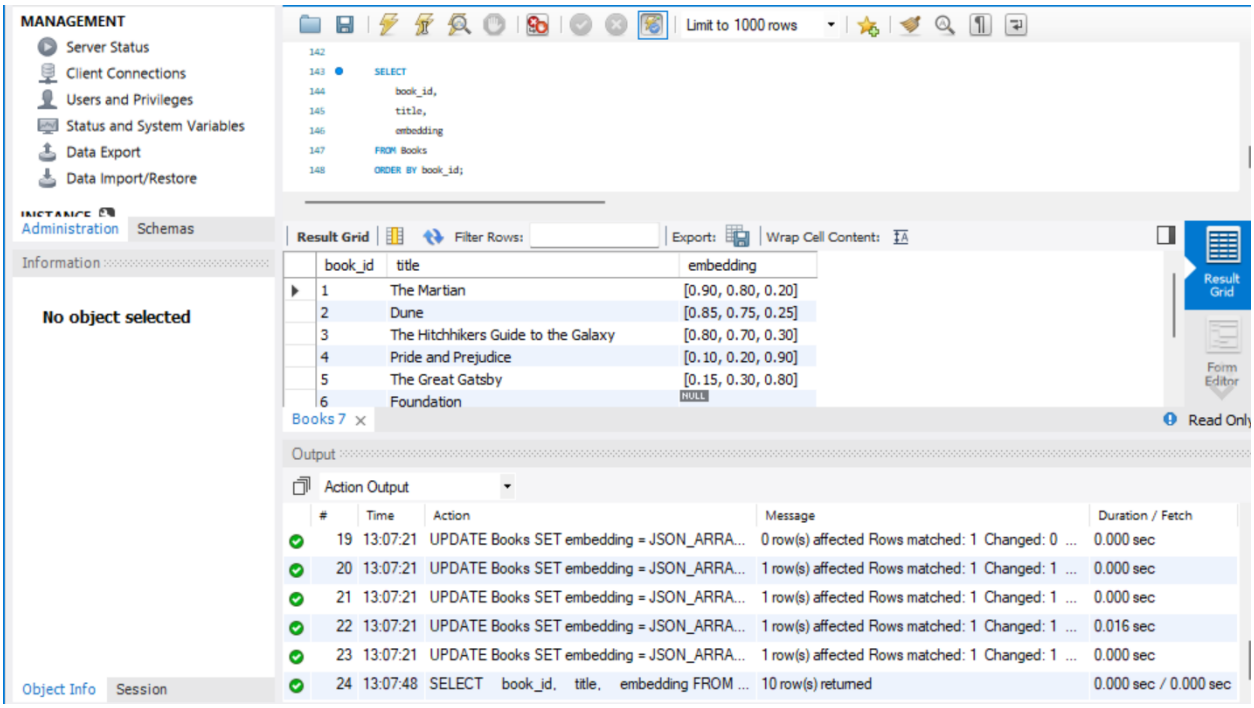

Administration
Schemas
Information
No object selected
Object Info
Session
'Twenty Thousand Leagues Under the Seas',
'Jules Verne',
'Adventure',
'A fantastic underwater adventure aboard a mysterious submarine.',
1870
);
SELECT * FROM Books;
ALTER TABLE Books
ADD COLUMN embedding JSON;
Output
Action Output
Time Action Message Duration / Fetch
5 13:00:01 CREATE TABLE Books (book_id INT AUTO_I... 0 row(s) affected 0.015 sec
6 13:01:04 DESCRIBE Books 6 row(s) returned 0.000 sec / 0.000 sec
7 13:01:42 title, author, genre, description, publishe... Error Code: 1064. You have an error in your SQL s... 0.000 sec
8 13:01:53 INSERT INTO Books (title, author, genre,... 10 row(s) affected Records: 10 Duplicates: 0 Wa... 0.000 sec
9 13:02:34 SELECT * FROM Books LIMIT 0, 1000 10 row(s) returned 0.000 sec / 0.000 sec
10 13:03:19 ALTER TABLE Books ADD COLUMN embedding... 0 row(s) affected Records: 0 Duplicates: 0 Warni... 0.016 sec

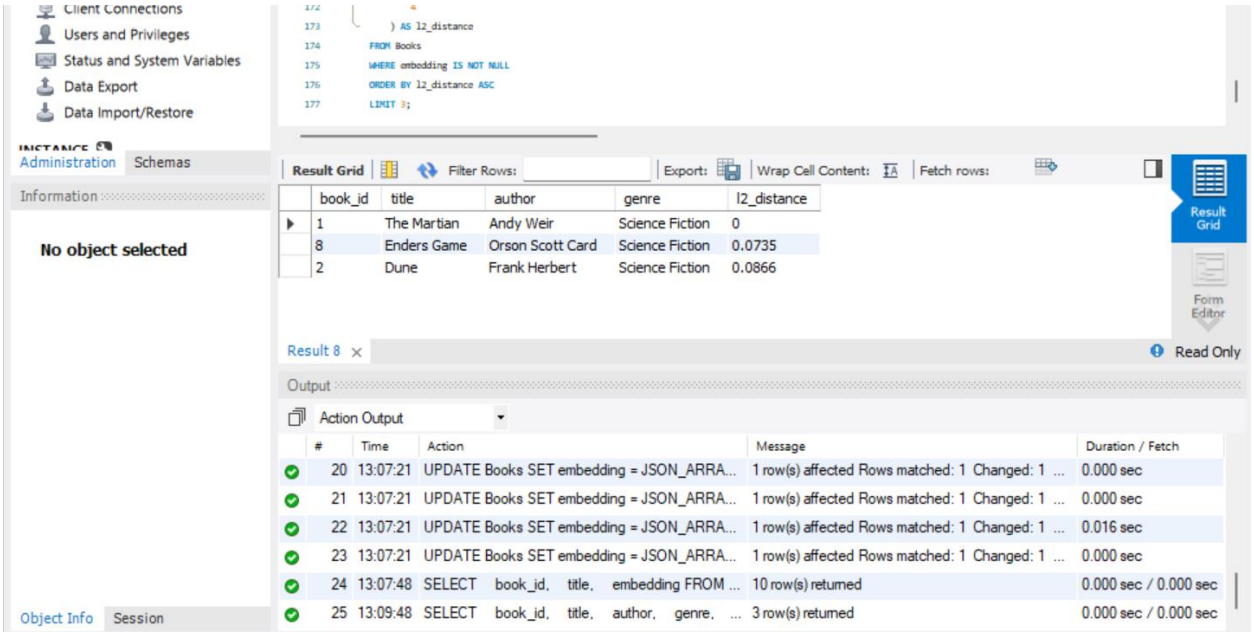

Data Export
Data Import/Restore
Administration
Schemas
Information
No object selected
Object Info
Session
DESCRIBE Books;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Field Type Null Key Default Extra
author varchar(150) NO NULL
genre varchar(100) YES NULL
description text YES NULL
published_year int YES NULL
embedding json YES NULL
Result 6
Output
Action Output
Time Action Message Duration / Fetch
6 13:01:04 DESCRIBE Books 6 row(s) returned 0.000 sec / 0.000 sec
7 13:01:42 title, author, genre, description, publishe... Error Code: 1064. You have an error in your SQL s... 0.000 sec
8 13:01:53 INSERT INTO Books (title, author, genre,... 10 row(s) affected Records: 10 Duplicates: 0 Wa... 0.000 sec
9 13:02:34 SELECT * FROM Books LIMIT 0, 1000 10 row(s) returned 0.000 sec / 0.000 sec
10 13:03:19 ALTER TABLE Books ADD COLUMN embedding... 0 row(s) affected Records: 0 Duplicates: 0 Warni... 0.016 sec
11 13:03:52 DESCRIBE Books 7 row(s) returned 0.000 sec / 0.000 sec


Information
No object selected
Object Info
Session
ALTER TABLE Books
ADD COLUMN embedding JSON;
DESCRIBE Books;
UPDATE Books
SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
WHERE book_id = 1;
Output
Action Output
Time Action Message Duration / Fetch
7 13:01:42 title, author, genre, description, publishe... Error Code: 1064. You have an error in your SQL s... 0.000 sec
8 13:01:53 INSERT INTO Books (title, author, genre,... 10 row(s) affected Records: 10 Duplicates: 0 Wa... 0.000 sec
9 13:02:34 SELECT * FROM Books LIMIT 0, 1000 10 row(s) returned 0.000 sec / 0.000 sec
10 13:03:19 ALTER TABLE Books ADD COLUMN embedding... 0 row(s) affected Records: 0 Duplicates: 0 Warni... 0.016 sec
11 13:03:52 DESCRIBE Books 7 row(s) returned 0.000 sec / 0.000 sec
12 13:04:30 UPDATE Books SET embedding = JSON_ARRA... 1 row(s) affected Rows matched: 1 Changed: 1 ... 0.000 sec


Information
No object selected
Object Info
Session
UPDATE Books
SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
WHERE book_id = 1;
UPDATE Books
SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
WHERE book_id = 2;
Output
Action Output
Time Action Message Duration / Fetch
8 13:01:53 INSERT INTO Books (title, author, genre,... 10 row(s) affected Records: 10 Duplicates: 0 Wa... 0.000 sec
9 13:02:34 SELECT * FROM Books LIMIT 0, 1000 10 row(s) returned 0.000 sec / 0.000 sec
10 13:03:19 ALTER TABLE Books ADD COLUMN embedding... 0 row(s) affected Records: 0 Duplicates: 0 Warni... 0.016 sec
11 13:03:52 DESCRIBE Books 7 row(s) returned 0.000 sec / 0.000 sec
12 13:04:30 UPDATE Books SET embedding = JSON_ARRA... 1 row(s) affected Rows matched: 1 Changed: 1 ... 0.000 sec
13 13:05:08 UPDATE Books SET embedding = JSON_ARRA... 1 row(s) affected Rows matched: 1 Changed: 1 ... 0.000 sec


MANAGEMENT
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Information
No object selected
Object Info
Session
Limit to 1000 rows
122
123 UPDATE Books
124 SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
125 WHERE book_id = 5;
126
127 UPDATE Books
128 SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
129 WHERE book_id = 7;
130
131 UPDATE Books
132 SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
133 WHERE book_id = 8;
134
135 UPDATE Books
136 SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)
137 WHERE book_id = 9;
138
139 UPDATE Books
140 SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)
Output
Action Output
Time Action Message Duration / Fetch
18 13:07:21 UPDATE Books SET embedding = JSON_ARRA... 0 row(s) affected Rows matched: 1 Changed: 0 ... 0.000 sec
19 13:07:21 UPDATE Books SET embedding = JSON_ARRA... 0 row(s) affected Rows matched: 1 Changed: 0 ... 0.000 sec
20 13:07:21 UPDATE Books SET embedding = JSON_ARRA... 1 row(s) affected Rows matched: 1 Changed: 1 ... 0.000 sec
21 13:07:21 UPDATE Books SET embedding = JSON_ARRA... 1 row(s) affected Rows matched: 1 Changed: 1 ... 0.000 sec
22 13:07:21 UPDATE Books SET embedding = JSON_ARRA... 1 row(s) affected Rows matched: 1 Changed: 1 ... 0.016 sec
23 13:07:21 UPDATE Books SET embedding = JSON_ARRA... 1 row(s) affected Rows matched: 1 Changed: 1 ... 0.000 sec


MANAGEMENT
Server Status
Client Connections
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Data Export
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Administration
Schemas
Information
No object selected
Object Info
Session
Limit to 1000 rows
142
143 SELECT
144 book_id,
145 title,
146 embedding
147 FROM Books
148 ORDER BY book_id;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
book_id title embedding
1 The Martian [0.90, 0.80, 0.20]
2 Dune [0.85, 0.75, 0.25]
3 The Hitchhikers Guide to the Galaxy [0.80, 0.70, 0.30]
4 Pride and Prejudice [0.10, 0.20, 0.90]
5 The Great Gatsby [0.15, 0.30, 0.80]
6 Foundation NULL
Books 7
Result Grid
Form Editor
Read Only
Output
Action Output
Time Action Message Duration / Fetch
19 13:07:21 UPDATE Books SET embedding = JSON_ARRA... 0 row(s) affected Rows matched: 1 Changed: 0 ... 0.000 sec
20 13:07:21 UPDATE Books SET embedding = JSON_ARRA... 1 row(s) affected Rows matched: 1 Changed: 1 ... 0.000 sec
21 13:07:21 UPDATE Books SET embedding = JSON_ARRA... 1 row(s) affected Rows matched: 1 Changed: 1 ... 0.000 sec
22 13:07:21 UPDATE Books SET embedding = JSON_ARRA... 1 row(s) affected Rows matched: 1 Changed: 1 ... 0.016 sec
23 13:07:21 UPDATE Books SET embedding = JSON_ARRA... 1 row(s) affected Rows matched: 1 Changed: 1 ... 0.000 sec
24 13:07:48 SELECT book_id, title, embedding FROM ... 10 row(s) returned 0.000 sec / 0.000 sec


Client Connections
Users and Privileges
Status and System Variables
Data Export
Data Import/Restore
Administration
Schemas
Information
No object selected
Object Info
Session
) AS l2_distance
FROM Books
WHERE embedding IS NOT NULL
ORDER BY l2_distance ASC
LIMIT 3;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Fetch rows:
book_id title author genre l2_distance
1 The Martian Andy Weir Science Fiction 0
8 Enders Game Orson Scott Card Science Fiction 0.0735
2 Dune Frank Herbert Science Fiction 0.0866
Result 8
Read Only
Output
Action Output
Time Action Message Duration / Fetch
20 13:07:21 UPDATE Books SET embedding = JSON_ARRA... 1 row(s) affected Rows matched: 1 Changed: 1 ... 0.000 sec
21 13:07:21 UPDATE Books SET embedding = JSON_ARRA... 1 row(s) affected Rows matched: 1 Changed: 1 ... 0.000 sec
22 13:07:21 UPDATE Books SET embedding = JSON_ARRA... 1 row(s) affected Rows matched: 1 Changed: 1 ... 0.016 sec
23 13:07:21 UPDATE Books SET embedding = JSON_ARRA... 1 row(s) affected Rows matched: 1 Changed: 1 ... 0.000 sec
24 13:07:48 SELECT book_id, title, embedding FROM ... 10 row(s) returned 0.000 sec / 0.000 sec
25 13:09:48 SELECT book_id, title, author, genre, ... 3 row(s) returned 0.000 sec / 0.000 sec